

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 06 • Forces

With Answers

Questions with full worked solutions

- 25 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 06

Forces

Your classified route

11

QUESTIONS

105

MARKS

2

FAMILIES

6

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 06 | FAMILY 01

Resolve forces for particle equilibrium

WHAT THIS FAMILY ASKS

Use a free-body diagram and two independent component equilibrium equations.

VARIANT ROUTE

V01 Horizontal/vertical equilibrium

V02 Ring, knot or multiple strings

3 classified questions

20 total marks

January 2022 Q1 — complete original question

1.

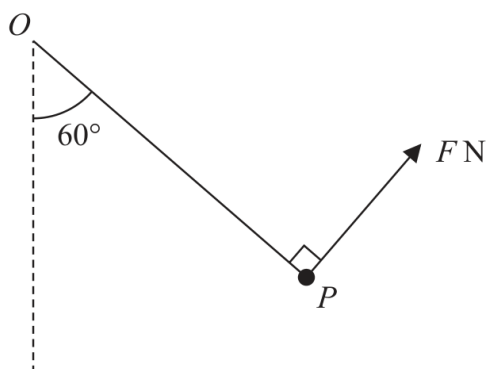


Figure 1

A particle P of weight 5 N is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O . The particle P is held in equilibrium by a force of magnitude F newtons. The direction of this force is perpendicular to the string and OP makes an angle of 60° with the vertical, as shown in Figure 1.

Find

(a) the value of F (3)

(b) the tension in the string. (3)

January 2022 Q1 — detailed solution

Component equilibrium

ALL
6 marks**Resolve perpendicular to the string**

The force F is perpendicular to the string. The weight is 5 N vertically downward, and the string makes 60° with the vertical.

Find F

The component of the weight perpendicular to the string is $5 \cos 30^\circ$, so $F = 5 \cos 30^\circ = \frac{5\sqrt{3}}{2}$ N.

Resolve along the string

The tension balances the component of the weight along the string. This component is $5 \sin 30^\circ$.

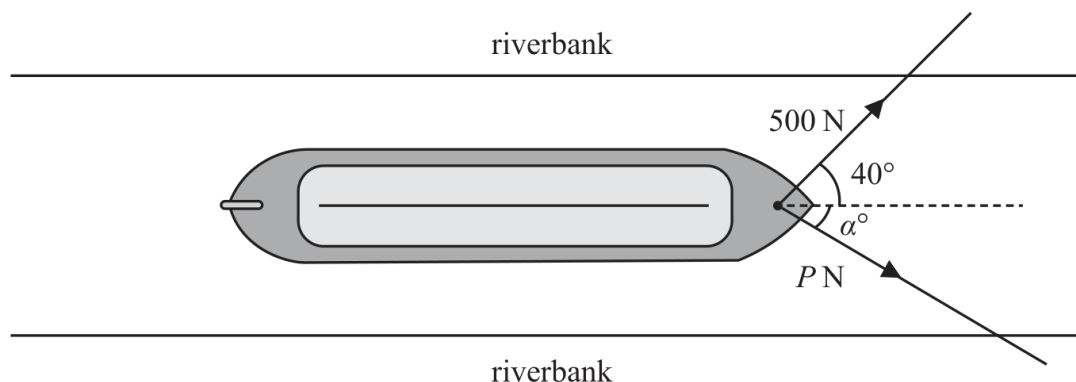
Find T

$$T = 5 \sin 30^\circ = \frac{5}{2} \text{ N.}$$

Final answer (a) $F = \frac{5\sqrt{3}}{2}$ N ≈ 4.3 N. (b) $T = \frac{5}{2}$ N.

January 2023 Q6 — complete original question

6.

**Figure 3**

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of 40° to the direction of motion, as shown in Figure 3.
- The tension in the second rope is P newtons, acting at an angle of α° to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- the value of α
- the value of P

(8)

January 2023 Q6 — detailed solution

Component equilibrium

ALL
8 marks**Resolve perpendicular to the motion**

The boat travels straight, so the sideways components balance: $500 \sin 40^\circ = P \sin \alpha$.

Resolve along the motion

The speed is constant, so the forward components balance the resistance: $500 \cos 40^\circ + P \cos \alpha = 900$.

Eliminate P

From the two equations, $P \sin \alpha = 500 \sin 40^\circ$ and $P \cos \alpha = 900 - 500 \cos 40^\circ$.

Find alpha

Divide the sine equation by the cosine equation: $\tan \alpha = \frac{500 \sin 40^\circ}{900 - 500 \cos 40^\circ}$. This gives $\alpha = 31.9^\circ$, so $\alpha = 32^\circ$ approximately.

Find P

Use $P \sin \alpha = 500 \sin 40^\circ$: $P = \frac{500 \sin 40^\circ}{\sin 31.9^\circ} = 608.7 \dots$ N. Hence $P \approx 610$ N.

Final answer (i) $\alpha = 32^\circ$ approximately. (ii) $P = 610$ N approximately.

January 2024 Q1 — complete original question

1.

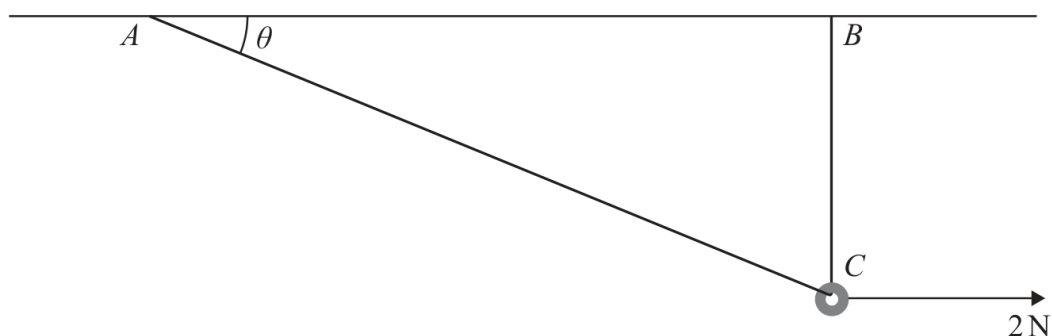


Figure 1

Figure 1 shows a small smooth ring **threaded** onto a light inextensible string.

One end of the string is attached to a fixed point A on a horizontal ceiling and the other end of the string is attached to a fixed point B on the ceiling.

A horizontal force of magnitude 2 N acts on the ring so that the ring rests in equilibrium at a point C , vertically below B , with the string taut.

The line of action of the horizontal force and the string both lie in the same vertical plane.

The angle that the string makes with the ceiling at A is θ , where $\tan \theta = \frac{3}{4}$

The tension in the string is T newtons. The mass of the ring is M kg.

(a) Find the value of T (3)

(b) Find the value of M (3)

January 2024 Q1 — detailed solution

Ring-and-string equilibrium

ALL
6 marks**Use the trig ratios**

Given $\tan \theta = \frac{3}{4}$, use $\sin \theta = \frac{3}{5}$ and $\cos \theta = \frac{4}{5}$.

Resolve horizontally

The horizontal force of 2 N is balanced by the horizontal component of the tension in the sloping part of the string: $T \cos \theta = 2$.

Find the tension

$T \cdot \frac{4}{5} = 2$, so $T = 2.5$ N.

Resolve vertically

The ring is smooth, so the tension is the same in both parts of the string. Upward components are T from the vertical part and $T \sin \theta$ from the sloping part. Therefore $T + T \sin \theta = Mg$.

Find the mass

$Mg = 2.5 + 2.5 \cdot \frac{3}{5} = 4.0$, so $M = \frac{4.0}{9.8} = 0.408$ kg.

Final answer (a) $T = 2.5$ N. (b) $M = 0.408$ kg approximately.

TOPIC 06 | FAMILY 02

Rough-contact equilibrium

WHAT THIS FAMILY ASKS

Solve static or limiting friction on non-incline rough contacts, with geometry made explicit at variant level: horizontal surface or vertical guide.

VARIANT ROUTE

V01 Actual versus limiting friction on a horizontal contact

V02 Single limiting threshold on a horizontal contact

V03 Vertical wire/rod minimum-maximum range

V04 Oblique force changes a horizontal-contact reaction

8 classified questions

85 total marks

Summer 2022 Q4 — complete original question

4.

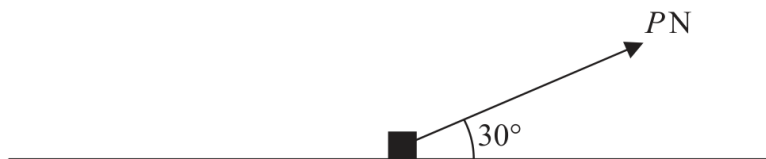


Figure 1

A small block of mass 5 kg lies at rest on a rough horizontal plane.

The coefficient of friction between the block and the plane is $\frac{3}{7}$

A force of magnitude P newtons is applied to the block in a direction which makes an angle of 30° with the plane, as shown in Figure 1.

The block is modelled as a particle.

Given that $P = 14$

- (a) find the magnitude of the frictional force exerted on the block by the plane and describe what happens to the block, justifying your answer.

(6)

The value of P is now changed so that the block is on the point of slipping along the plane.

- (b) Find the value of P

(6)

Summer 2022 Q4 — detailed solution

Actual versus limiting friction

ALL
12 marks**Find the normal reaction when $P = 14$** Resolve vertically. The upward component of the applied force is $14 \sin 30^\circ$, so $R + 14 \sin 30^\circ = 5g$.**Calculate R** $R + 7 = 49$, so $R = 42$ N.**Find the maximum friction**

$$F_{\max} = \mu R = \frac{3}{7}(42) = 18 \text{ N.}$$

Compare with the horizontal pullThe horizontal component of the applied force is $14 \cos 30^\circ = 12.1 \dots$ N. Since this is less than the maximum possible friction, the block does not move.**State the actual friction**For equilibrium, the actual friction equals the horizontal pull, so $F = 12.1$ N approximately.**Set up the limiting case**When the block is on the point of slipping, friction is limiting. Resolve vertically: $R + P \sin 30^\circ = 5g$, so $R = 5g - P \sin 30^\circ$.**Resolve horizontally**At the limiting case, $P \cos 30^\circ = \frac{3}{7}R = \frac{3}{7}(5g - P \sin 30^\circ)$.**Find P** Substitute $g = 9.8$, $\sin 30^\circ = \frac{1}{2}$: $P \cos 30^\circ = \frac{3}{7}(49 - \frac{1}{2}P)$. Solving gives $P = 19.4 \dots$ N.**Final answer** (a) $F = 12.1$ N approximately, and the block does not move. (b) $P = 19.4$ N approximately.

January 2020 Q4 — complete original question

4.

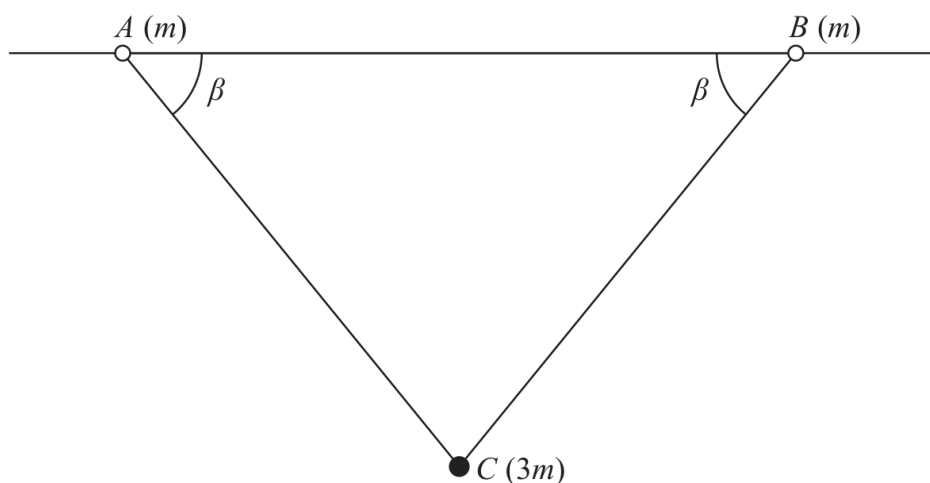


Figure 2

Two identical small rings, A and B , each of mass m , are threaded onto a rough horizontal wire. The rings are connected by a light inextensible string. A particle C of mass $3m$ is attached to the midpoint of the string. The particle C hangs in equilibrium below the wire with angle $BAC = \beta$, as shown in Figure 2.

The tension in each of the parts, AC and BC , of the string is T

(a) By considering particle C , find T in terms of m , g and β (2)

(b) Find, in terms of m and g , the magnitude of the normal reaction between the wire and A . (3)

The coefficient of friction between each ring and the wire is $\frac{4}{5}$

The two rings, A and B , are on the point of sliding along the wire towards each other.

(c) Find the value of $\tan \beta$ (5)

January 2020 Q4 — detailed solution

Horizontal limiting threshold

ALL
10 marks**Resolve forces on C**

The two tensions are equal and symmetric. Their vertical components support the weight of C, so $2T \sin \beta = 3mg$. Hence $T = \frac{3mg}{2 \sin \beta}$.

Resolve vertically for ring A

For ring A, the normal reaction balances its own weight and the downward vertical component of the string tension: $R = mg + T \sin \beta$.

Find the normal reaction

From $2T \sin \beta = 3mg$, we have $T \sin \beta = \frac{3}{2}mg$. Therefore $R = mg + \frac{3}{2}mg = \frac{5}{2}mg$.

Use limiting friction

The rings are just about to slide towards each other, so friction is limiting. With $\mu = \frac{4}{5}$, $F = \mu R = \frac{4}{5} \cdot \frac{5}{2}mg = 2mg$.

Resolve horizontally for a ring

The horizontal component of the tension is balanced by friction: $T \cos \beta = F = 2mg$.

Find tan beta

Substitute $T = \frac{3mg}{2 \sin \beta}$: $\frac{3mg}{2 \sin \beta} \cos \beta = 2mg$. Hence $\frac{3}{2} \cot \beta = 2$, so $\cot \beta = \frac{4}{3}$ and $\tan \beta = \frac{3}{4}$.

Final answer (a) $T = \frac{3mg}{2 \sin \beta}$. (b) $R = \frac{5}{2}mg$. (c) $\tan \beta = \frac{3}{4}$.

Summer 2023 Q6 — complete original question

6.

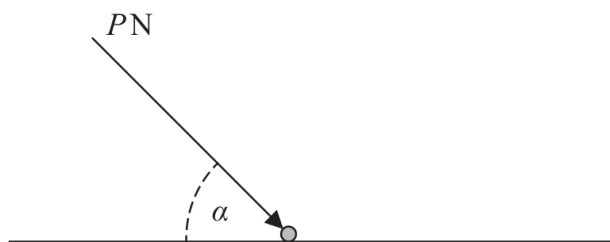


Figure 3

A particle of weight W newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude P newtons is applied to the particle.

The force acts at an angle α to the horizontal, where $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is $\frac{1}{4}$

Given that the particle does not move, show that

$$P \leq \frac{5W}{8}$$

(7)

Summer 2023 Q6 — detailed solution

Horizontal limiting threshold

ALL
7 marks**Use the trig ratios**

Given $\tan \alpha = \frac{4}{3}$, use $\sin \alpha = \frac{4}{5}$ and $\cos \alpha = \frac{3}{5}$.

Resolve vertically

The applied force increases the normal reaction in this model, so $R = W + P \sin \alpha$.

Resolve horizontally

For the particle not to move, friction must be able to balance the horizontal component of the applied force, so $P \cos \alpha \leq F_{\max}$.

Use limiting friction

Since $\mu = \frac{1}{4}$, $F_{\max} = \frac{1}{4}R = \frac{1}{4}(W + P \sin \alpha)$.

Substitute the exact trig values

Thus $P \cdot \frac{3}{5} \leq \frac{1}{4} \left(W + P \cdot \frac{4}{5} \right)$.

Rearrange the inequality

Multiplying by 20 gives $12P \leq 5W + 4P$. Hence $8P \leq 5W$, so $P \leq \frac{5}{8}W$.

Final answer $P \leq \frac{5}{8}W$.

October 2021 Q5 — complete original question

5.

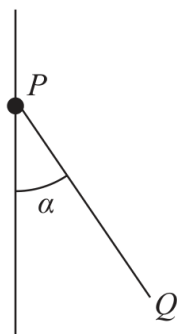


Figure 2

A small bead of mass 0.2 kg is attached to the end P of a light rod PQ . The bead is threaded onto a fixed vertical rough wire.

The bead is held in equilibrium with the rod PQ inclined to the wire at an angle α , where $\tan \alpha = \frac{4}{3}$, as shown in Figure 2.

The thrust in the rod is T newtons.

The bead is modelled as a particle.

- (a) Find the magnitude and direction of the friction force acting on the bead when $T = 2.5$
- (3)

The coefficient of friction between the bead and the wire is μ .

Given that the greatest possible value of T is 6.125

- (b) find the value of μ .
- (7)

October 2021 Q5 — detailed solution

Vertical-guide minimum and maximum range

ALL
10 marks**Use the trig ratios**

Since $\tan \alpha = \frac{4}{3}$, $\sin \alpha = \frac{4}{5}$ and $\cos \alpha = \frac{3}{5}$.

Resolve vertically when $T = 2.5$

The vertical component of the thrust is $2.5 \cos \alpha = 2.5 \cdot \frac{3}{5} = 1.5 \text{ N}$ upward. The weight is $0.2g = 1.96 \text{ N}$ downward.

Find friction and direction

For equilibrium, friction must supply the missing upward force: $F = 1.96 - 1.5 = 0.46 \text{ N}$. So friction acts upwards.

Use greatest T

When $T = 6.125$, the vertical component of the thrust is larger than the weight, so friction acts downward at the limiting value.

Find normal reaction

The normal reaction is the horizontal component of the thrust: $R = 6.125 \sin \alpha = 6.125 \cdot \frac{4}{5} = 4.9 \text{ N}$.

Find limiting friction

Vertical equilibrium gives $0.2g + F = 6.125 \cos \alpha$. Thus $1.96 + F = 6.125 \cdot \frac{3}{5} = 3.675$, so $F = 1.715 \text{ N}$.

Find μ

$F = \mu R$, so $\mu = \frac{1.715}{4.9} = 0.35$.

Final answer (a) $F = 0.46 \text{ N}$ upwards. (b) $\mu = 0.35$.

October 2023 Q5 — complete original question

5.

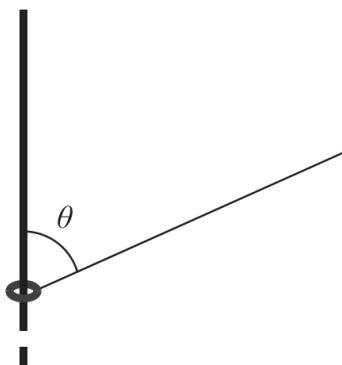


Figure 3

A small ring of mass 0.2 kg is attached to one end of a light inextensible string.

The ring is **threaded** onto a fixed rough vertical rod.

The string is taut and makes an angle θ with the rod, as shown in Figure 3, where $\tan \theta = \frac{12}{5}$

Given that the ring is in equilibrium and that the tension in the string is 10 N ,

(a) find the magnitude of the frictional force acting on the ring,

(3)

(b) state the direction of the frictional force acting on the ring.

(1)

The coefficient of friction between the ring and the rod is $\frac{1}{4}$

Given that the ring is in equilibrium, and that the tension in the string, T newtons, can now vary,

(c) (i) find the minimum value of T

(ii) find the maximum value of T

(8)

October 2023 Q5 — detailed solution

Vertical-guide minimum and maximum range

ALL
12 marks**Use the trig ratios**

The string makes angle θ with the vertical rod and $\tan \theta = \frac{12}{5}$. Hence $\sin \theta = \frac{12}{13}$ and $\cos \theta = \frac{5}{13}$.

Find the friction for $T = 10$

The upward component of the tension is $10 \cos \theta = 10 \cdot \frac{5}{13} = \frac{50}{13}$ N. The weight is $0.2g = 1.96$ N.

Balance vertical forces

Since the upward component of tension is larger than the weight, friction must act downwards. Its magnitude is $\frac{50}{13} - 1.96 = 1.89 \dots$ N, so $F = 1.9$ N.

Write the normal reaction

For variable tension T , the normal reaction from the rod is produced by the horizontal component of tension: $R = T \sin \theta = \frac{12T}{13}$.

Use limiting friction

Since $\mu = \frac{1}{4}$, limiting friction is $F = \mu R = \frac{1}{4} \cdot \frac{12T}{13} = \frac{3T}{13}$.

Find the minimum tension

For the minimum tension, the ring would tend to slide down, so friction acts upwards. Vertical equilibrium gives $T \cos \theta + F = 0.2g$, hence $\frac{5T}{13} + \frac{3T}{13} = 1.96$. Thus $T = 3.185 \dots$ N, so $T_{\min} = 3.2$ N.

Find the maximum tension

For the maximum tension, the ring would tend to move up, so friction acts downwards. Vertical equilibrium gives $T \cos \theta = 0.2g + F$, hence $\frac{5T}{13} = 1.96 + \frac{3T}{13}$. Thus $T = 12.74 \dots$ N, so $T_{\max} = 13$ N.

Final answer (a) $F = 1.9$ N. (b) Friction acts downwards. (c)(i) $T_{\min} = 3.2$ N. (c)(ii) $T_{\max} = 13$ N.

January 2021 Q3 — complete original question

3.

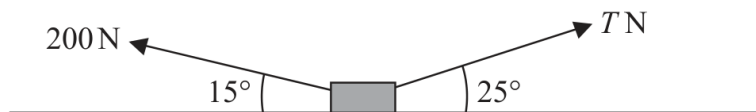


Figure 1

A parcel of mass 20 kg is at rest on a rough horizontal floor. The coefficient of friction between the parcel and the floor is 0.3

Two forces, both acting in the same vertical plane, of magnitudes 200 N and $T\text{ N}$ are applied to the parcel. The line of action of the 200 N force makes an angle of 15° with the horizontal and the line of action of the $T\text{ N}$ force makes an angle of 25° with the horizontal, as shown in Figure 1. The parcel is modelled as a particle P .

Find the smallest value of T for which P remains in equilibrium.

(9)

January 2021 Q3 — detailed solution

Oblique forces on rough horizontal contact

ALL
9 marks**Resolve vertically**For vertical equilibrium, $R + 200 \sin 15^\circ + T \sin 25^\circ = 20g$.**Resolve horizontally**For the smallest T , friction is limiting and balances the remaining horizontal tendency. Horizontally, $200 \cos 15^\circ - T \cos 25^\circ - F = 0$.**Use limiting friction** $F = 0.3R$.**Eliminate R and F**From the vertical equation, $R = 20g - 200 \sin 15^\circ - T \sin 25^\circ$. Substitute into $F = 0.3R$ and then into the horizontal equation.**Solve for T** $200 \cos 15^\circ - T \cos 25^\circ - 0.3(20g - 200 \sin 15^\circ - T \sin 25^\circ) = 0$. Solving gives $T = 192.3 \dots$ N, so the smallest value is about 192 N.**Final answer** $T = 192$ N approximately.

Summer 2024 Q6 — complete original question

6.

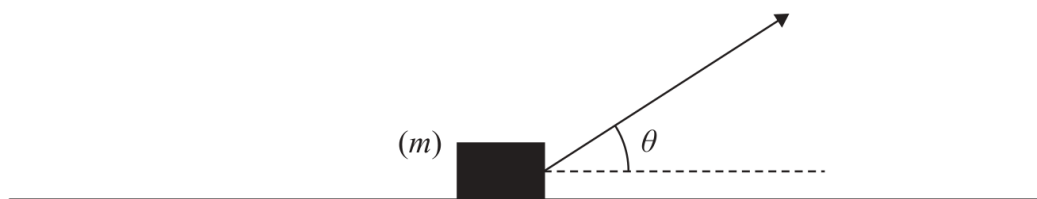


Figure 3

A box of mass m lies on a rough horizontal plane. The box is pulled along the plane in a straight line at **constant speed** by a light rope. The rope is inclined at an angle θ to the plane, as shown in Figure 3.

The coefficient of friction between the box and the plane is $\frac{1}{3}$

The box is modelled as a particle.

Given that $\tan \theta = \frac{3}{4}$

(a) find, in terms of m and g , the tension in the rope.

(7)

The rope is now removed and the box is placed at rest on the plane. The box is then projected horizontally along the plane with speed u .

The box is again modelled as a particle.

When the box has moved a distance d along the plane, the speed of the box is $\frac{1}{2}u$.

(b) Find d in terms of u and g .

(5)

Summer 2024 Q6 — detailed solution

Oblique forces on rough horizontal contact

ALL
12 marks**Use the trig ratios**Given $\tan \theta = \frac{3}{4}$, use $\sin \theta = \frac{3}{5}$ and $\cos \theta = \frac{4}{5}$.**Resolve vertically**The box moves at constant speed, so it is in equilibrium. Vertically, $R + T \sin \theta = mg$, hence $R = mg - T \sin \theta$.**Resolve horizontally**Horizontally, the tension component balances friction: $T \cos \theta = F$.**Use friction**With $\mu = \frac{1}{3}$, $F = \frac{1}{3}R$. Therefore $T \cos \theta = \frac{1}{3}(mg - T \sin \theta)$.**Find T**Substitute $\sin \theta = \frac{3}{5}$ and $\cos \theta = \frac{4}{5}$: $\frac{4}{5}T = \frac{1}{3}\left(mg - \frac{3}{5}T\right)$. This gives $\frac{4}{5}T = \frac{mg}{3} - \frac{1}{5}T$, so $T = \frac{mg}{3}$.**Find the deceleration when the rope is removed**On the horizontal plane without the rope, $R = mg$ and friction is $\frac{1}{3}mg$. Hence the acceleration is $-\frac{g}{3}$.**Use SUVAT to find d**When the speed has fallen from u to $\frac{1}{2}u$, use $v^2 = u^2 + 2as$: $\left(\frac{u}{2}\right)^2 = u^2 + 2\left(-\frac{g}{3}\right)d$.**Solve for d** $\frac{u^2}{4} = u^2 - \frac{2gd}{3}$, so $\frac{2gd}{3} = \frac{3u^2}{4}$ and $d = \frac{9u^2}{8g}$.**Final answer** (a) $T = \frac{mg}{3}$. (b) $d = \frac{9u^2}{8g}$.

October 2025 Q5 — complete original question

5.

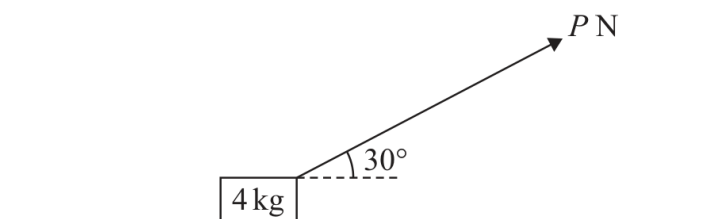


Figure 3

A box of mass 4 kg is placed on a rough horizontal surface.

A force of magnitude P newtons, acting at 30° to the horizontal, is applied to the box, as shown in Figure 3.

The coefficient of friction between the box and the surface is $\frac{2}{3}$

The box is modelled as a particle.

(a) Find the value of P when the box is on the point of sliding along the surface. (6)

The value of P is now increased to 25 and the box moves along the surface.

Find

(b) the acceleration of the box, (5)

(c) the speed of the box when it has moved 1.5 m. (2)

October 2025 Q5 — detailed solution

Oblique forces on rough horizontal contact

ALL
13 marks**Part (a): Resolve vertically**

$$R + P \sin 30^\circ = 4g, \text{ so } R = 39.2 - \frac{P}{2}.$$

Part (a): Use limiting friction

$$\text{At the point of sliding, } F = \mu R = \frac{2}{3} \left(39.2 - \frac{P}{2} \right).$$

Part (a): Resolve horizontally

$$P \cos 30^\circ = F. \text{ Hence } \frac{\sqrt{3}}{2} P = \frac{2}{3} \left(39.2 - \frac{P}{2} \right), \text{ giving } P = 21.8 \text{ N to 3 significant figures.}$$

Part (b): Find the normal reaction

$$\text{When } P = 25 \text{ N, } R = 39.2 - \frac{25}{2} = 26.7 \text{ N.}$$

Part (b): Find friction and acceleration

$$F = \frac{2}{3}(26.7) = 17.8 \text{ N. Horizontally, } 25 \cos 30^\circ - 17.8 = 4a, \text{ so } a = 0.9625 \text{ m s}^{-2} \approx 0.96 \text{ m s}^{-2}.$$

Part (c): Apply constant-acceleration kinematics

$$\text{The box starts from rest. With } u = 0, a = 0.9625 \text{ m s}^{-2} \text{ and } s = 1.5 \text{ m, use } v^2 = u^2 + 2as. \text{ Thus } v = \sqrt{2(0.9625)(1.5)} = 1.699 \dots \text{ m s}^{-1}.$$

Final answer (a) $P = 21.8 \text{ N}$. (b) $a = 0.96 \text{ m s}^{-2}$. (c) $v = 1.7 \text{ m s}^{-1}$.